

## WEEK 5

Divyadharshini G

# HEALTHCARE DATASET PREPARATION AND DATA MODELING

## 1. Introduction

The objective of Week 5 was to prepare the healthcare dataset using Power Query, create separate dimension tables, generate Patient ID and Admission ID, and create a Billing table for data modeling in Power BI.

## 2. Data Cleaning

The healthcare dataset was imported into Power BI and opened in Power Query for data preparation. The categorical text columns were cleaned using **Trim** and **Clean** transformations to remove unnecessary spaces and unwanted characters. The **Name** column was standardized using **Capitalize Each Word** to maintain consistent capitalization.

Table.AddColumn(#"Cleaned Text5", "Age Group", each if [Age] < 18 then "Child" else if [Age] <= 30 then "Young Adult" else if [Age] <=

|    | Valid               | Error | Empty | Valid | Error  | Empty | Valid | Error        | Empty | Valid      | Error         | Empty | Valid | Error | Empty | Valid | Error | Empty | Valid | Error | Empty |
|----|---------------------|-------|-------|-------|--------|-------|-------|--------------|-------|------------|---------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1  | Bobby Jackson       |       |       | 30    | Male   |       | B-    | Cancer       |       | 31-01-2024 | Matthew Smit  |       |       |       |       |       |       |       |       |       |       |
| 2  | Leslie Terry        |       |       | 62    | Male   |       | A+    | Obesity      |       | 20-08-2019 | Samantha Da   |       |       |       |       |       |       |       |       |       |       |
| 3  | Danny Smith         |       |       | 76    | Female |       | A-    | Obesity      |       | 22-09-2022 | Tiffany Mitch |       |       |       |       |       |       |       |       |       |       |
| 4  | Andrew Watts        |       |       | 28    | Female |       | O+    | Diabetes     |       | 18-11-2020 | Kevin Wells   |       |       |       |       |       |       |       |       |       |       |
| 5  | Adrienne Bell       |       |       | 43    | Female |       | AB+   | Cancer       |       | 19-09-2022 | Kathleen Han  |       |       |       |       |       |       |       |       |       |       |
| 6  | Emily Johnson       |       |       | 36    | Male   |       | A+    | Asthma       |       | 20-12-2023 | Taylor Newto  |       |       |       |       |       |       |       |       |       |       |
| 7  | Edward Edwards      |       |       | 21    | Female |       | AB-   | Diabetes     |       | 03-11-2020 | Kelly Olson   |       |       |       |       |       |       |       |       |       |       |
| 8  | Christina Martinez  |       |       | 20    | Female |       | A+    | Cancer       |       | 28-12-2021 | Suzanne Thor  |       |       |       |       |       |       |       |       |       |       |
| 9  | Jasmine Aguilar     |       |       | 82    | Male   |       | AB+   | Asthma       |       | 01-07-2020 | Daniel Fergus |       |       |       |       |       |       |       |       |       |       |
| 10 | Christopher Berg    |       |       | 58    | Female |       | AB-   | Cancer       |       | 23-05-2021 | Heather Day   |       |       |       |       |       |       |       |       |       |       |
| 11 | Michelle Daniels    |       |       | 72    | Male   |       | O+    | Cancer       |       | 19-04-2020 | John Duncan   |       |       |       |       |       |       |       |       |       |       |
| 12 | Aaron Martinez      |       |       | 38    | Female |       | A-    | Hypertension |       | 13-08-2023 | Douglas Mayn  |       |       |       |       |       |       |       |       |       |       |
| 13 | Connor Hansen       |       |       | 75    | Female |       | A+    | Diabetes     |       | 12-12-2019 | Kenneth Fletc |       |       |       |       |       |       |       |       |       |       |
| 14 | Robert Bauer        |       |       | 68    | Female |       | AB+   | Asthma       |       | 22-05-2020 | Theresa Free  |       |       |       |       |       |       |       |       |       |       |
| 15 | Brooke Brady        |       |       | 44    | Female |       | AB+   | Cancer       |       | 08-10-2021 | Roberta Stew  |       |       |       |       |       |       |       |       |       |       |
| 16 | Ms. Natalie Gamble  |       |       | 46    | Female |       | AB-   | Obesity      |       | 01-01-2023 | Maria Doughi  |       |       |       |       |       |       |       |       |       |       |
| 17 | Haley Perkins       |       |       | 63    | Female |       | A+    | Arthritis    |       | 23-06-2020 | Erica Spencer |       |       |       |       |       |       |       |       |       |       |
| 18 | Mrs. Jamie Campbell |       |       | 38    | Male   |       | AB-   | Obesity      |       | 08-03-2020 | Justin Kim    |       |       |       |       |       |       |       |       |       |       |
| 19 | Luke Burgess        |       |       | 34    | Female |       | A-    | Hypertension |       | 04-03-2021 | Justin Moore  |       |       |       |       |       |       |       |       |       |       |
| 20 | Daniel Schmidt      |       |       | 63    | Male   |       | B+    | Asthma       |       | 15-11-2022 | Denise Gallov |       |       |       |       |       |       |       |       |       |       |
| 21 | Timothy Burns       |       |       | 67    | Female |       | A-    | Asthma       |       | 28-06-2023 | Krista Smith  |       |       |       |       |       |       |       |       |       |       |
| 22 | Christopher Bright  |       |       | 48    | Male   |       | B+    | Asthma       |       | 21-01-2020 | Gregory Smit  |       |       |       |       |       |       |       |       |       |       |
| 23 | Kathryn Stewart     |       |       | 58    | Female |       | O+    | Arthritis    |       | 12-05-2022 | Vanessa New   |       |       |       |       |       |       |       |       |       |       |
| 24 | Dr. Eileen Thompson |       |       | 59    | Male   |       | A+    | Asthma       |       | 02-08-2021 | Donna Marti   |       |       |       |       |       |       |       |       |       |       |
| 25 | Dr. Eileen Thompson |       |       | 59    | Male   |       | A+    | Asthma       |       | 02-08-2021 | Donna Marti   |       |       |       |       |       |       |       |       |       |       |

Query Settings

PROPERTIES

Name

healthcare\_dataset

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Trimmed Text
- Cleaned Text
- Capitalized Each Word
- Trimmed Text1
- Cleaned Text2
- Trimmed Text2
- Trimmed Text3
- Cleaned Text3
- Trimmed Text4
- Cleaned Text4
- Trimmed Text5
- Cleaned Text5
- Added Conditional Column

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## 3. Creating Age Group

An Age Group conditional column was created based on the patient's age.

The age groups used were:

- 0–18
- 19–35
- 36–60
- 60+

This helps in analyzing patients according to different age ranges. Using conditional column.

×

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

Age Group

|         | Column Name | Operator                 | Value ⓘ      |      | Output ⓘ                 |
|---------|-------------|--------------------------|--------------|------|--------------------------|
| If      | Age ▾       | is less than ▾           | ABC 123 ▾ 18 | Then | ABC 123 ▾ Child          |
| Else If | Age ▾       | is less than or equ... ▾ | ABC 123 ▾ 30 | Then | ABC 123 ▾ Young Adult    |
| Else If | Age ▾       | is less than or equ... ▾ | ABC 123 ▾ 50 | Then | ABC 123 ▾ Adult          |
| Else If | Age ▾       | is less than or equ... ▾ | ABC 123 ▾ 65 | Then | ABC 123 ▾ Middle Age ... |

Add Clause

Else ⓘ

ABC 123 ▾ Senior

OK

Cancel

## 4. Creating Dim\_Patient

A separate Dim\_Patient table was created by keeping patient-related information.

| Queries [3]        |                     |                  |        |                  |             |                  |            |
|--------------------|---------------------|------------------|--------|------------------|-------------|------------------|------------|
| healthcare_dataset | Dim_Patient         | Dim_Admission    |        |                  |             |                  |            |
| A <sup>1</sup> C   | Name                | Y <sup>2</sup> 3 | Age    | A <sup>1</sup> C | Gender      | A <sup>1</sup> C | Blood Type |
| Valid              | 100%                | Valid            | 100%   | Valid            | 100%        | Valid            | 100%       |
| Error              | 0%                  | Error            | 0%     | Error            | 0%          | Error            | 0%         |
| Empty              | 0%                  | Empty            | 0%     | Empty            | 0%          | Empty            | 0%         |
| 1                  | Bobby Jackson       | 30               | Male   | B-               | Young Adult |                  |            |
| 2                  | Leslie Terry        | 62               | Male   | A+               | Middle Age  |                  |            |
| 3                  | Danny Smith         | 76               | Female | A-               | Senior      |                  |            |
| 4                  | Andrew Watts        | 28               | Female | O+               | Young Adult |                  |            |
| 5                  | Adrienne Bell       | 43               | Female | AB+              | Adult       |                  |            |
| 6                  | Emily Johnson       | 36               | Male   | A+               | Adult       |                  |            |
| 7                  | Edward Edwards      | 21               | Female | AB-              | Young Adult |                  |            |
| 8                  | Christina Martinez  | 20               | Female | A+               | Young Adult |                  |            |
| 9                  | Jasmine Aguilar     | 82               | Male   | AB+              | Senior      |                  |            |
| 10                 | Christopher Berg    | 58               | Female | AB-              | Middle Age  |                  |            |
| 11                 | Michelle Daniels    | 72               | Male   | O+               | Senior      |                  |            |
| 12                 | Aaron Martinez      | 38               | Female | A-               | Adult       |                  |            |
| 13                 | Connor Hansen       | 75               | Female | A+               | Senior      |                  |            |
| 14                 | Robert Bauer        | 68               | Female | AB+              | Senior      |                  |            |
| 15                 | Brooke Brady        | 44               | Female | AB+              | Adult       |                  |            |
| 16                 | Ms. Natalie Gamble  | 46               | Female | AB-              | Adult       |                  |            |
| 17                 | Haley Perkins       | 63               | Female | A+               | Middle Age  |                  |            |
| 18                 | Mrs. Jamie Campbell | 38               | Male   | AB-              | Adult       |                  |            |
| 19                 | Luke Burgess        | 34               | Female | A-               | Adult       |                  |            |
| 20                 | Daniel Schmidt      | 63               | Male   | B+               | Middle Age  |                  |            |
| 21                 | Timothy Burns       | 67               | Female | A-               | Senior      |                  |            |
| 22                 | Christopher Bright  | 48               | Male   | B+               | Adult       |                  |            |
| 23                 | Kathryn Stewart     | 58               | Female | O+               | Middle Age  |                  |            |
| 24                 | Dr. Eileen Thompson | 59               | Male   | A+               | Middle Age  |                  |            |
| 25                 | Paul Henderson      | 72               | Female | AB+              | Senior      |                  |            |

## 5. Creating Patient ID

An **Index Column** was added to the Dim\_Patient table and renamed as **Patient ID**.Patient ID provides an identifier that can be used to connect patient information with the Billing table.

| Queries [4]        |                     |                  |         |                  |             |                  |            |
|--------------------|---------------------|------------------|---------|------------------|-------------|------------------|------------|
| healthcare_dataset | Dim_Admission       | Dim_patient      | Billing |                  |             |                  |            |
| A <sup>1</sup> C   | Name                | Y <sup>2</sup> 3 | Age     | A <sup>1</sup> C | Gender      | A <sup>1</sup> C | Blood Type |
| Valid              | 100%                | Valid            | 100%    | Valid            | 100%        | Valid            | 100%       |
| Error              | 0%                  | Error            | 0%      | Error            | 0%          | Error            | 0%         |
| Empty              | 0%                  | Empty            | 0%      | Empty            | 0%          | Empty            | 0%         |
| 1                  | Bobby Jackson       | 30               | Male    | B-               | Young Adult | 1                |            |
| 2                  | Leslie Terry        | 62               | Male    | A+               | Middle Age  | 2                |            |
| 3                  | Danny Smith         | 76               | Female  | A-               | Senior      | 3                |            |
| 4                  | Andrew Watts        | 28               | Female  | O+               | Young Adult | 4                |            |
| 5                  | Adrienne Bell       | 43               | Female  | AB+              | Adult       | 5                |            |
| 6                  | Emily Johnson       | 36               | Male    | A+               | Adult       | 6                |            |
| 7                  | Edward Edwards      | 21               | Female  | AB-              | Young Adult | 7                |            |
| 8                  | Christina Martinez  | 20               | Female  | A+               | Young Adult | 8                |            |
| 9                  | Jasmine Aguilar     | 82               | Male    | AB+              | Senior      | 9                |            |
| 10                 | Christopher Berg    | 58               | Female  | AB-              | Middle Age  | 10               |            |
| 11                 | Michelle Daniels    | 72               | Male    | O+               | Senior      | 11               |            |
| 12                 | Aaron Martinez      | 38               | Female  | A-               | Adult       | 12               |            |
| 13                 | Connor Hansen       | 75               | Female  | A+               | Senior      | 13               |            |
| 14                 | Robert Bauer        | 68               | Female  | AB+              | Senior      | 14               |            |
| 15                 | Brooke Brady        | 44               | Female  | AB+              | Adult       | 15               |            |
| 16                 | Ms. Natalie Gamble  | 46               | Female  | AB-              | Adult       | 16               |            |
| 17                 | Haley Perkins       | 63               | Female  | A+               | Middle Age  | 17               |            |
| 18                 | Mrs. Jamie Campbell | 38               | Male    | AB-              | Adult       | 18               |            |
| 19                 | Luke Burgess        | 34               | Female  | A-               | Adult       | 19               |            |
| 20                 | Daniel Schmidt      | 63               | Male    | B+               | Middle Age  | 20               |            |
| 21                 | Timothy Burns       | 67               | Female  | A-               | Senior      | 21               |            |
| 22                 | Christopher Bright  | 48               | Male    | B+               | Adult       | 22               |            |
| 23                 | Kathryn Stewart     | 58               | Female  | O+               | Middle Age  | 23               |            |
| 24                 | Dr. Eileen Thompson | 59               | Male    | A+               | Middle Age  | 24               |            |
| 25                 | Paul Henderson      | 72               | Female  | AB+              | Senior      | 25               |            |

## 6. Creating Dim\_Admission

A separate Dim\_Admission table was created to store admission-related information. Admission ID , Admission Type, Doctor ,Hospital ,Medical Condition, Medication ,Room Number Test Results.

| Insurance Provider | Admission Type | Discharge Date | Medication  | Test Results | Admission ID |
|--------------------|----------------|----------------|-------------|--------------|--------------|
| Blue Cross         | Urgent         | 27-10-2023     | Penicillin  | Normal       | 77           |
| Cigna              | Emergency      | 04-02-2021     | Penicillin  | Inconclusive | 78           |
| UnitedHealthcare   | Emergency      | 05-12-2021     | Penicillin  | Inconclusive | 79           |
| UnitedHealthcare   | Urgent         | 26-03-2024     | Ibuprofen   | Inconclusive | 80           |
| Blue Cross         | Emergency      | 08-12-2020     | Ibuprofen   | Abnormal     | 81           |
| Aetna              | Urgent         | 27-08-2020     | Penicillin  | Abnormal     | 82           |
| Blue Cross         | Elective       | 12-12-2023     | Penicillin  | Abnormal     | 83           |
| Cigna              | Emergency      | 14-06-2020     | Aspirin     | Inconclusive | 84           |
| Medicare           | Emergency      | 10-06-2023     | Ibuprofen   | Abnormal     | 85           |
| UnitedHealthcare   | Emergency      | 23-07-2022     | Ibuprofen   | Normal       | 86           |
| Aetna              | Emergency      | 08-10-2020     | Penicillin  | Inconclusive | 87           |
| UnitedHealthcare   | Emergency      | 25-08-2023     | Ibuprofen   | Inconclusive | 88           |
| Cigna              | Elective       | 01-06-2023     | Aspirin     | Abnormal     | 89           |
| Blue Cross         | Elective       | 07-09-2022     | Paracetamol | Inconclusive | 90           |
| Medicare           | Elective       | 28-06-2022     | Penicillin  | Abnormal     | 91           |
| UnitedHealthcare   | Emergency      | 25-11-2022     | Lipitor     | Abnormal     | 92           |
| Blue Cross         | Elective       | 18-09-2021     | Ibuprofen   | Inconclusive | 93           |
| Cigna              | Urgent         | 27-09-2022     | Ibuprofen   | Inconclusive | 94           |
| Aetna              | Elective       | 10-05-2021     | Lipitor     | Inconclusive | 95           |
| Medicare           | Elective       | 26-08-2019     | Lipitor     | Abnormal     | 96           |
| Aetna              | Elective       | 14-04-2024     | Aspirin     | Abnormal     | 97           |
| UnitedHealthcare   | Elective       | 31-12-2022     | Ibuprofen   | Normal       | 98           |
| Blue Cross         | Elective       | 18-09-2022     | Aspirin     | Normal       | 99           |
| UnitedHealthcare   | Elective       | 11-11-2022     | Aspirin     | Abnormal     | 100          |

## 7. Creating Billing Table

A separate Billing table was created from the prepared healthcare data.

The required billing and admission-related fields were retained for further modeling.

|    |     | Admission Type                     | Discharge Date                     | Medication                         | Test Results                       | Age Group                          | Dim_patient.patient ID             |
|----|-----|------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
|    |     | 100% Valid<br>0% Error<br>0% Empty | 100% Valid<br>0% Error<br>0% Empty | 100% Valid<br>0% Error<br>0% Empty | 100% Valid<br>0% Error<br>0% Empty | 100% Valid<br>0% Error<br>0% Empty | 100% Valid<br>0% Error<br>0% Empty |
| 1  | 328 | Urgent                             | 02-02-2024                         | Paracetamol                        | Normal                             | Young Adult                        |                                    |
| 2  | 265 | Emergency                          | 26-08-2019                         | Ibuprofen                          | Inconclusive                       | Middle Age                         |                                    |
| 3  | 205 | Emergency                          | 07-10-2022                         | Aspirin                            | Normal                             | Senior                             |                                    |
| 4  | 450 | Elective                           | 18-12-2020                         | Ibuprofen                          | Abnormal                           | Young Adult                        |                                    |
| 5  | 458 | Urgent                             | 09-10-2022                         | Penicillin                         | Abnormal                           | Adult                              |                                    |
| 6  | 389 | Urgent                             | 24-12-2023                         | Ibuprofen                          | Normal                             | Adult                              |                                    |
| 7  | 389 | Emergency                          | 15-11-2020                         | Paracetamol                        | Inconclusive                       | Young Adult                        |                                    |
| 8  | 277 | Emergency                          | 07-01-2022                         | Paracetamol                        | Inconclusive                       | Young Adult                        |                                    |
| 9  | 316 | Elective                           | 14-07-2020                         | Aspirin                            | Abnormal                           | Senior                             |                                    |
| 10 | 249 | Elective                           | 22-06-2021                         | Paracetamol                        | Inconclusive                       | Middle Age                         |                                    |
| 11 | 394 | Urgent                             | 22-04-2020                         | Paracetamol                        | Normal                             | Senior                             |                                    |
| 12 | 288 | Urgent                             | 05-09-2023                         | Lipitor                            | Inconclusive                       | Adult                              |                                    |
| 13 | 134 | Emergency                          | 28-12-2019                         | Penicillin                         | Abnormal                           | Senior                             |                                    |
| 14 | 309 | Urgent                             | 19-06-2020                         | Lipitor                            | Normal                             | Senior                             |                                    |
| 15 | 182 | Urgent                             | 13-10-2021                         | Paracetamol                        | Normal                             | Adult                              |                                    |
| 16 | 465 | Elective                           | 11-01-2023                         | Aspirin                            | Inconclusive                       | Adult                              |                                    |
| 17 | 114 | Elective                           | 14-07-2020                         | Paracetamol                        | Normal                             | Middle Age                         |                                    |
| 18 | 449 | Urgent                             | 02-04-2020                         | Paracetamol                        | Abnormal                           | Adult                              |                                    |
| 19 | 260 | Elective                           | 14-03-2021                         | Aspirin                            | Abnormal                           | Adult                              |                                    |
| 20 | 465 | Elective                           | 22-11-2022                         | Penicillin                         | Normal                             | Middle Age                         |                                    |
| 21 | 115 | Elective                           | 02-07-2023                         | Aspirin                            | Normal                             | Senior                             |                                    |
| 22 | 295 | Urgent                             | 09-02-2020                         | Lipitor                            | Normal                             | Adult                              |                                    |

## 8.Merging Billing with Dim\_Patient

The Billing table was merged with **Dim\_Patient** using matching patient information.A **Left Outer Join** was used. After the merge, **Patient ID** was selected and expanded into the Billing table.

Close & Apply

New Source

Recent Sources

Enter Data

Data source settings

Manage Parameters

Export query results

Refresh Preview

Properties Advanced Editor

Choose Columns

Remove Columns

Keep Rows

Remove Rows

Sort

Split Column

Group By

Data Type: Whole Number

Use First Row as Headers

Replace Values

Merge Queries

Append Queries

Combine Files

Combine

Queries [4]

healthcare\_dataset

Dim\_Admission

Dim\_patient

Billing

Table.ExpandTableColumn("#Merged Queries", "Dim\_patient", ("patient ID"), ("Dim\_patient.patient ID"))

|    |     | Admission Type | Discharge Date | Medication  | Test Results | Age Group   | Dim_patient.patient ID |
|----|-----|----------------|----------------|-------------|--------------|-------------|------------------------|
| 1  | 328 | Urgent         | 02-02-2024     | Paracetamol | Normal       | Young Adult |                        |
| 2  | 265 | Emergency      | 26-08-2019     | Ibuprofen   | Inconclusive | Middle Age  |                        |
| 3  | 205 | Emergency      | 07-10-2022     | Aspirin     | Normal       | Senior      |                        |
| 4  | 450 | Elective       | 18-12-2020     | Ibuprofen   | Abnormal     | Young Adult |                        |
| 5  | 458 | Urgent         | 09-10-2022     | Penicillin  | Abnormal     | Adult       |                        |
| 6  | 389 | Urgent         | 24-12-2023     | Ibuprofen   | Normal       | Adult       |                        |
| 7  | 389 | Emergency      | 15-11-2020     | Paracetamol | Inconclusive | Young Adult |                        |
| 8  | 277 | Emergency      | 07-01-2022     | Paracetamol | Inconclusive | Young Adult |                        |
| 9  | 316 | Elective       | 14-07-2020     | Aspirin     | Abnormal     | Senior      |                        |
| 10 | 249 | Elective       | 22-06-2021     | Paracetamol | Inconclusive | Middle Age  |                        |
| 11 | 394 | Urgent         | 22-04-2020     | Paracetamol | Normal       | Senior      |                        |
| 12 | 288 | Urgent         | 05-09-2023     | Lipitor     | Inconclusive | Adult       |                        |
| 13 | 134 | Emergency      | 28-12-2019     | Penicillin  | Abnormal     | Senior      |                        |
| 14 | 309 | Urgent         | 19-06-2020     | Lipitor     | Normal       | Senior      |                        |
| 15 | 182 | Urgent         | 13-10-2021     | Paracetamol | Normal       | Adult       |                        |
| 16 | 465 | Elective       | 11-01-2023     | Aspirin     | Inconclusive | Adult       |                        |
| 17 | 114 | Elective       | 14-07-2020     | Paracetamol | Normal       | Middle Age  |                        |
| 18 | 449 | Urgent         | 02-04-2020     | Paracetamol | Abnormal     | Adult       |                        |
| 19 | 260 | Elective       | 14-03-2021     | Aspirin     | Abnormal     | Adult       |                        |
| 20 | 465 | Elective       | 22-11-2022     | Penicillin  | Normal       | Middle Age  |                        |
| 21 | 115 | Elective       | 02-07-2023     | Aspirin     | Normal       | Senior      |                        |
| 22 | 295 | Urgent         | 09-02-2020     | Lipitor     | Normal       | Adult       |                        |

PROPERTIES

Name

Billing

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

Cleaned Text

Capitalized Each Word

Trimmed Text1

Trimmed Text2

Cleaned Text2

Trimmed Text3

Cleaned Text3

Trimmed Text4

Cleaned Text4

Trimmed Text5

Cleaned Text5

Added Conditional Column

Merge

Select a table and matching columns to create a merged table.

Billing

| Name          | 1  | Age    | 2  | Gender   | 3          | Blood Type       | 4        | Medical Condition | Date of Admission | Doctor |  |
|---------------|----|--------|----|----------|------------|------------------|----------|-------------------|-------------------|--------|--|
| Bobby Jackson | 30 | Male   | B- | Cancer   | 31-01-2024 | Matthew Smith    | Sons and |                   |                   |        |  |
| Leslie Terry  | 62 | Male   | A+ | Obesity  | 20-08-2019 | Samantha Davies  | Kim Inc  |                   |                   |        |  |
| Danny Smith   | 76 | Female | A- | Obesity  | 22-09-2022 | Tiffany Mitchell | Cook PLC |                   |                   |        |  |
| Andrew Watts  | 28 | Female | O+ | Diabetes | 18-11-2020 | Kevin Wells      | Hernand  |                   |                   |        |  |

Dim\_patient

| Name          | 1  | Age    | 2   | Gender      | 3 | Blood Type | 4 | Age Group | 5 | patient ID |
|---------------|----|--------|-----|-------------|---|------------|---|-----------|---|------------|
| Bobby Jackson | 30 | Male   | B-  | Young Adult |   |            |   |           |   | 1          |
| Leslie Terry  | 62 | Male   | A+  | Middle Age  |   |            |   |           |   | 2          |
| Danny Smith   | 76 | Female | A-  | Senior      |   |            |   |           |   | 3          |
| Andrew Watts  | 28 | Female | O+  | Young Adult |   |            |   |           |   | 4          |
| Adrienne Bell | 43 | Female | AB+ | Adult       |   |            |   |           |   | 5          |

Join Kind

Left Outer (all from first, matching from second)

Use fuzzy matching to perform the merge

Fuzzy matching options

The selection matches 55500 of 55500 rows from the first table.

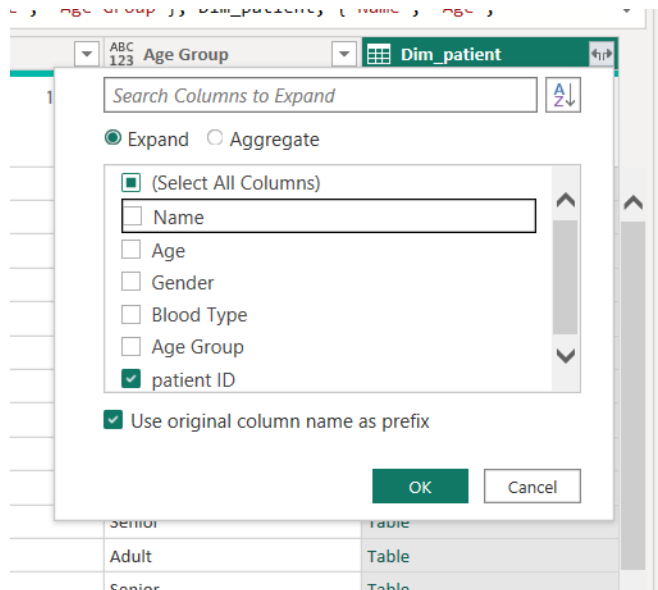
OK

Cancel

16 COLUMNS, 999+ ROWS

Column profiling based on top 1000 rows

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## 9. Merging Billing with Dim\_Admission

The Billing table was then merged with **Dim\_Admission** using admission information. A **Left Outer Join** was used. The **Admission ID** was selected from the merged table and expanded into Billing.

A **Left Outer Join** was used. The **Admission ID** was selected from the merged table and expanded into Billing.

## Merge

Select a table and matching columns to create a merged table.

Billing

| Name          | Age | Gender | Blood Type | Medical Condition | 1 | Date of Admission | 2 | Doctor           |
|---------------|-----|--------|------------|-------------------|---|-------------------|---|------------------|
| Bobby Jackson | 30  | Male   | B-         | Cancer            |   | 31-01-2024        |   | Matthew Smith    |
| Leslie Terry  | 62  | Male   | A+         | Obesity           |   | 20-08-2019        |   | Samantha Davies  |
| Danny Smith   | 76  | Female | A-         | Obesity           |   | 22-09-2022        |   | Tiffany Mitchell |
| Andrew Watts  | 28  | Female | O+         | Diabetes          |   | 18-11-2020        |   | Kevin Wells      |

Refresh

Dim\_Admission

| Medical Condition | 1 | Date of Admission | 2 | Doctor           | Hospital                   | Insurance Provider | Admission Type |
|-------------------|---|-------------------|---|------------------|----------------------------|--------------------|----------------|
| Cancer            |   | 31-01-2024        |   | Matthew Smith    | Sons and Miller            | Blue Cross         | Urgent         |
| Obesity           |   | 20-08-2019        |   | Samantha Davies  | Kim Inc                    | Medicare           | Emergency      |
| Obesity           |   | 22-09-2022        |   | Tiffany Mitchell | Cook PLC                   | Aetna              | Emergency      |
| Diabetes          |   | 18-11-2020        |   | Kevin Wells      | Hernandez Rogers and Vang, | Medicare           | Elective       |

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

▸ Fuzzy matching options

OK

Cancel

, "Doctor", "Hospital", "Insurance Provider",

Dim\_patient.patient ID Dim\_Admission

Search Columns to Expand

☒ Expand ☐ Aggregate

☒ (Select All Columns)

☒ Medical Condition

☐ Date of Admission

☐ Doctor

☐ Hospital

☐ Insurance Provider

☐ Admission Type

☐ Discharge Date

☐ Medication

☐ Test Results

☒ Admission ID

☒ Use original column name as prefix

OK Cancel

15 Table

16 Table

17 Table

18 Table

19 Table

20 Table

## 10. Final Billing Table

The final Billing table contains the main fields required for analysis: Date of Admission, Insurance Provider, Billing Amount, Discharge Date, Patient ID, Admission ID.

|    | Date of Admission | Insurance Provider | Billing Amount | Discharge Date | Dim_patient.patient ID | Dim_Admission.Admission ID |
|----|-------------------|--------------------|----------------|----------------|------------------------|----------------------------|
| 1  | 31-01-2024        | Blue Cross         | 18856.28131    | 02-02-2024     | 1                      | 1                          |
| 2  | 20-08-2019        | Medicare           | 33643.32729    | 26-08-2019     | 2                      | 2                          |
| 3  | 22-09-2022        | Aetna              | 27955.09608    | 07-10-2022     | 3                      | 3                          |
| 4  | 18-11-2020        | Medicare           | 37909.78241    | 18-12-2020     | 4                      | 4                          |
| 5  | 19-09-2022        | Aetna              | 14238.31781    | 09-10-2022     | 5                      | 5                          |
| 6  | 20-12-2023        | UnitedHealthcare   | 48145.11095    | 24-12-2023     | 6                      | 6                          |
| 7  | 03-11-2020        | Medicare           | 19580.87234    | 15-11-2020     | 7                      | 7                          |
| 8  | 28-12-2021        | Cigna              | 45820.46272    | 07-01-2022     | 8                      | 8                          |
| 9  | 01-07-2020        | Cigna              | 50119.22279    | 14-07-2020     | 9                      | 9                          |
| 10 | 23-05-2021        | UnitedHealthcare   | 19784.63106    | 22-06-2021     | 10                     | 10                         |
| 11 | 19-04-2020        | Medicare           | 12576.79561    | 22-04-2020     | 11                     | 11                         |
| 12 | 13-08-2023        | Medicare           | 7999.58688     | 05-09-2023     | 12                     | 12                         |
| 13 | 12-12-2019        | Cigna              | 43282.28336    | 28-12-2019     | 13                     | 13                         |
| 14 | 22-05-2020        | UnitedHealthcare   | 33207.70663    | 19-06-2020     | 14                     | 14                         |
| 15 | 08-10-2021        | UnitedHealthcare   | 40703.59923    | 13-10-2021     | 15                     | 15                         |
| 16 | 01-01-2023        | Blue Cross         | 12363.35743    | 11-01-2023     | 16                     | 16                         |
| 17 | 23-06-2020        | UnitedHealthcare   | 24499.8479     | 14-07-2020     | 17                     | 17                         |
| 18 | 08-09-2020        | Cigna              | 17440.46544    | 02-04-2020     | 18                     | 18                         |
| 19 | 04-03-2021        | Blue Cross         | 18843.02302    | 14-03-2021     | 19                     | 19                         |
| 20 | 15-11-2022        | Cigna              | 23762.20358    | 22-11-2022     | 20                     | 20                         |
| 21 | 28-06-2023        | Blue Cross         | 42.51458855    | 02-07-2023     | 21                     | 21                         |
| 22 | 21-01-2020        | Aetna              | 17695.91162    | 09-02-2020     | 22                     | 22                         |
| 23 | 12-05-2022        | Aetna              | 5998.102908    | 10-06-2022     | 23                     | 23                         |
| 24 | 02-08-2021        | Aetna              | 25250.05243    | 12-08-2021     | 24                     | 24                         |
| 25 | 15-05-2020        | Medicare           | 53313.50543    | 05-05-2020     | 25                     | 25                         |

## 11. Final Data Model

The final data model consists of: Dim\_Patient, Dim\_Admission, Billing.

